MYERS-STEENROD FOR COMPACT MANIFOLDS

SERKAN SALIK

The goal of this note is to provide a short proof of the fact that $\mathrm{Isom}(M)$ is a Lie group for a compact Riemannian manifold (M, g). The idea was found using ChatGPT 5.6 Sol.

Let $G = \mathrm{Isom}(M, g)$ with the compact-open topology. This is immediately a topological group because $\mathrm{Homeo}(X)$ is a topological group for compact Hausdorff spaces X. G is compact by Arzela-Ascoli: all isometries are 1-Lipschitz, so any sequence of them is equicontinuous, and clearly the limit of a sequence of isometries will be an isometry.

Now consider the operator $P := -\Delta + 1$. It decomposes $L^2(M)$ into finite dimensional eigenspaces $\bigoplus_\lambda E_\lambda$. G acts isometrically on each of these eigenspaces, since the Laplacian is a natural operator. Thus if we can show, for some large Λ that G acts faithfully on $\bigoplus_{\lambda \leq \Lambda} E_\lambda$, we get a compact and hence closed subgroup of the orthogonal group of $\bigoplus_{\lambda \leq \Lambda} E_\lambda$, hence a Lie group by the closed subgroup theorem.

To show this action is faithful, we'll show that there exists an embedding $\phi \colon M \to \mathbb{R}^N$, where the components of ϕ are P-eigenfunctions. In that case, if $g \in G$ acted trivially, then $g^*\phi = \phi$, but since ϕ is injective, we have $g = \mathrm{Id}$, so G is faithful. To show such an embedding exists, it suffices to show the P-eigenfunctions span a dense subset of C^1, since we can take an embedding, and any other map that is C^1-close enough to it will also be an embedding (in this case it will also be smooth, since P-eigenfunctions are smooth)

To that end, recall that P is invertible as an operator from $H^{k+2} \to H^k$ by Lax-Milgram and elliptic regularity. Since the eigenfunctions of P are dense in L^2, we get from invertibility of $-\Delta + 1 \colon H^2 \to L^2$ that they are dense in H^2. Iterating, they are dense in H^k for all k, and hence in C^k for all k, by Morrey embedding. Explicitly, if $u \in C^\infty$, it is in some Sobolev space H^ℓ for some large ℓ. In H^ℓ, we may approximate it to arbitrary precision by a finite linear combination v of P-eigenfunctions, which will be smooth and hence lie in any Sobolev space. Then under the Morrey embedding, they will be close in whichever C^k spaces H^ℓ embeds into. Since C^∞ functions are dense in C^k, and C^∞ functions are contained in the closure of the span of the P-eigenfunctions, we get density in C^k for all k, as required.